

Pulse code Modulation (PCM)

(Tx)

inf
Signal

LPF

Sampler

Quantizing

Encoder

PCM
Signal
101010

discretization

anti aliasing
effect

digital

Uniform

Non-uniform

Adaptive

compander

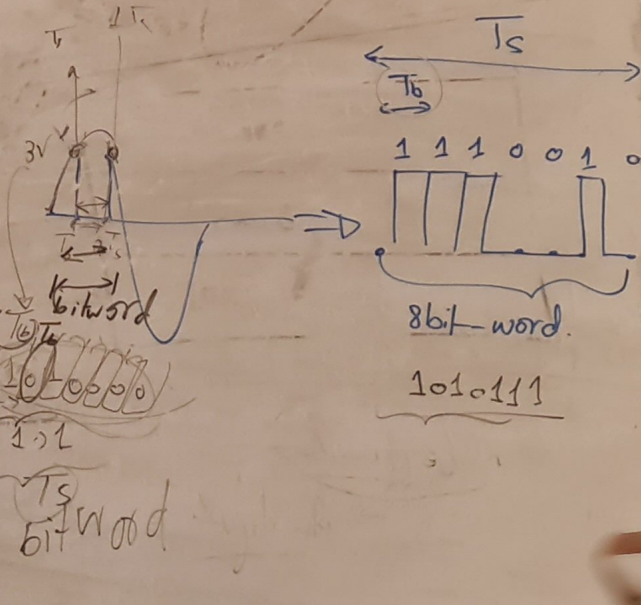
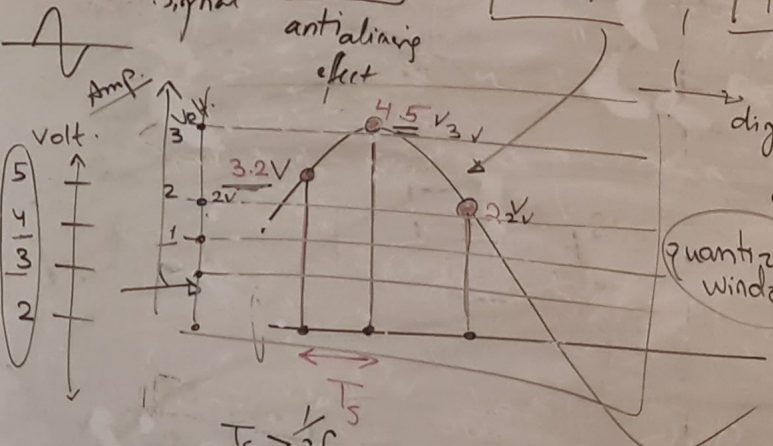
discretization in amp (quantizing)

+
discretization in time (sample)

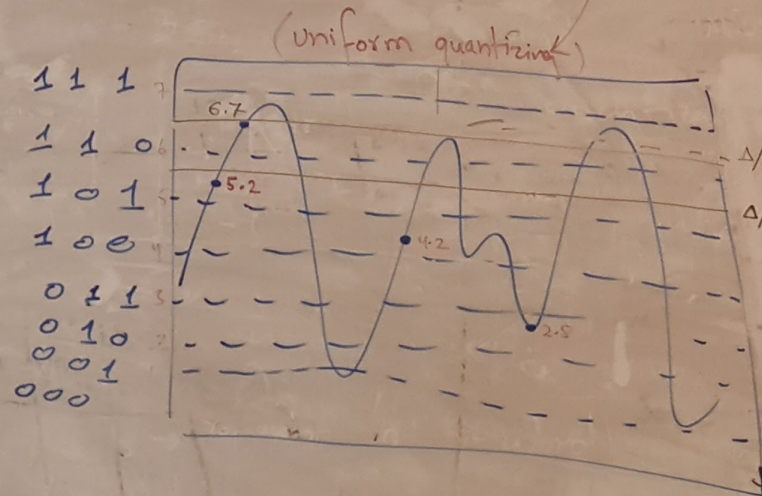
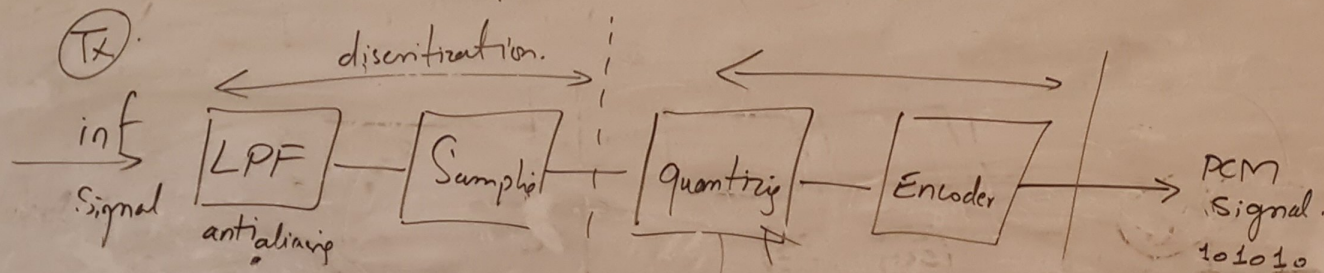
digital
form

$$T_s \geq \frac{1}{2f_m}$$

$T_s \rightarrow$ represent in bit word



Pulse code Modulation (PCM)

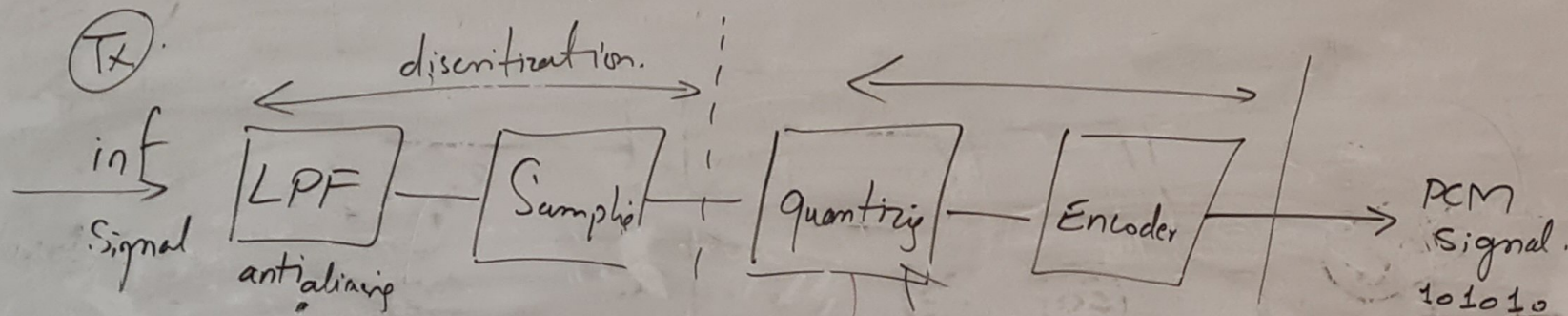


$$\Delta = \frac{8 - 0}{8} = 1 \text{ v.}$$

$$L = 2^R = 2^3 \rightarrow \text{No. of levels.}$$

$R \rightarrow \text{No of bits.}$

Pulse code Modulation (PCM)



8 levels

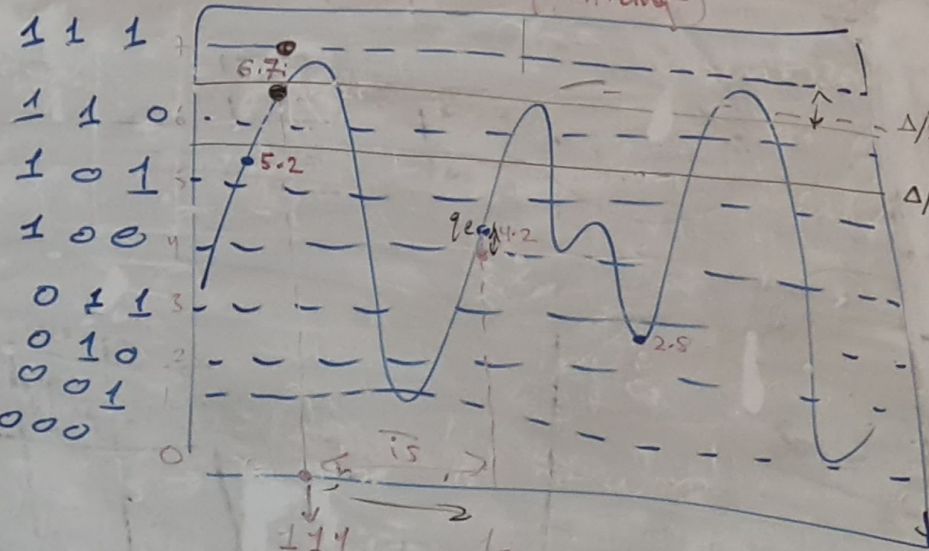
(uniform quantizing)

6.7 5 3

$$\Delta = \frac{8 - 0}{8} = 1v. \text{ Step size.}$$

$$L = 2^R = 2^3 \rightarrow \text{No. of levels.}$$

$R \rightarrow \text{No of bits.}$

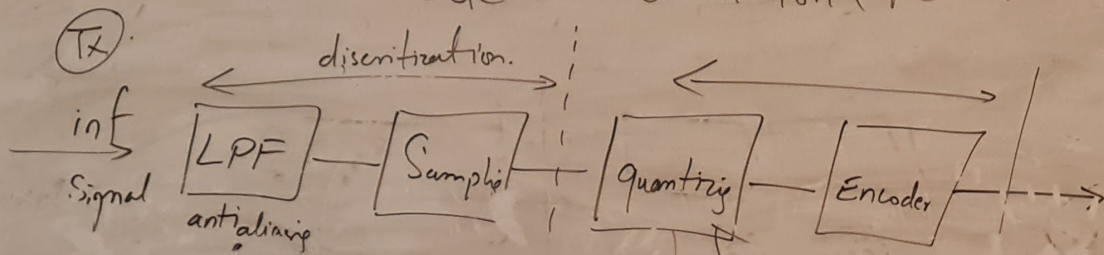


$$q = \frac{x_{inf}}{L} = \frac{7 - 6.7}{2} = 0.3$$

111 100 11

$\frac{137}{24}$

Pulse code Modulation (PCM)



quantization
Noise
power
level:

$$N_q = \frac{m_p^2}{3L^2}$$

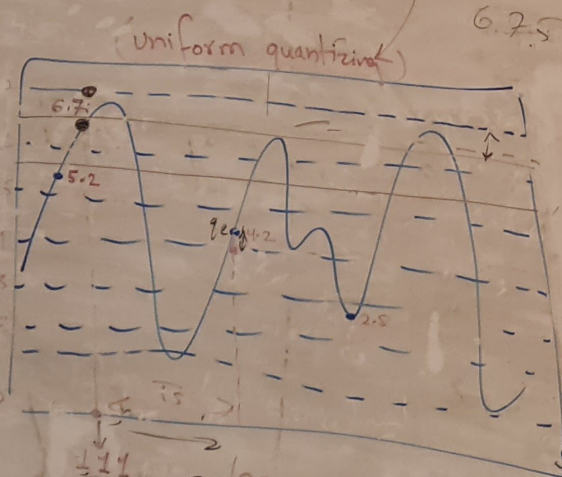
$$SQNR = \frac{3L^2 P}{m_p^2}$$

Signal to quantization power ratio.

$$R_b = R f_s = \frac{R}{T_s}$$

bitrate.

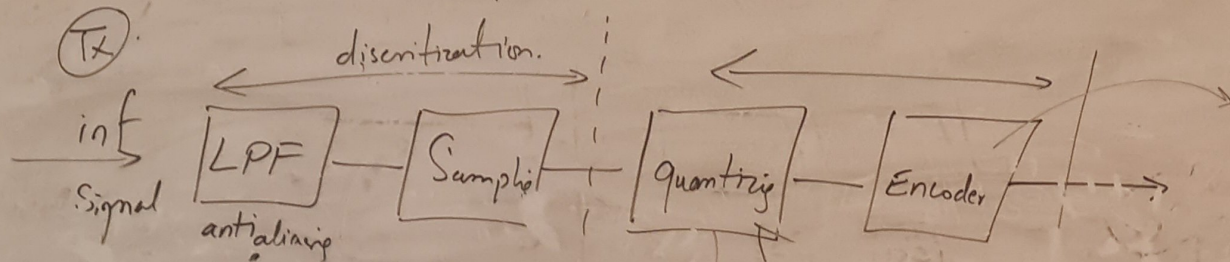
8 levels
 $q = \frac{7 - 6.7}{8} = 0.3$
 111
 110
 101
 100
 011
 010
 001
 000



111 100 011
 stream of bits

111 100 011
 111 100 011

Pulse code Modulation (PCM)



Line codes used for representing binary data stream

Unipolar NRZ / RZ

Polar NRZ / RZ

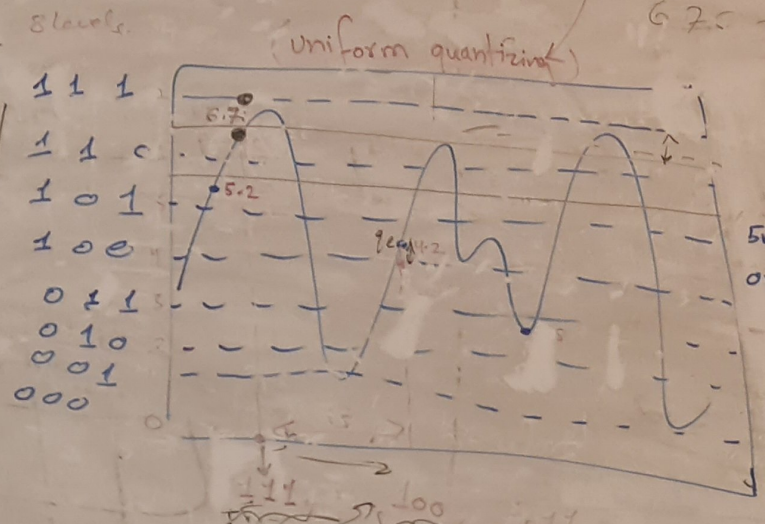
Bipolar (BRZ)

Split-phase signaling (Manchester Code)

x_{inf}

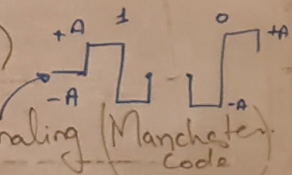
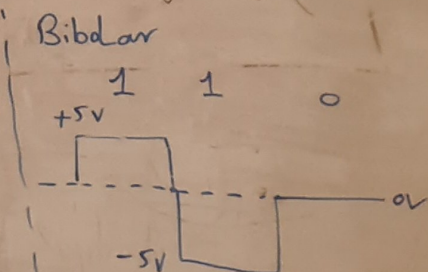
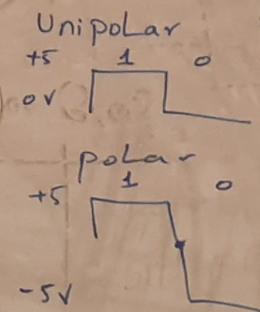
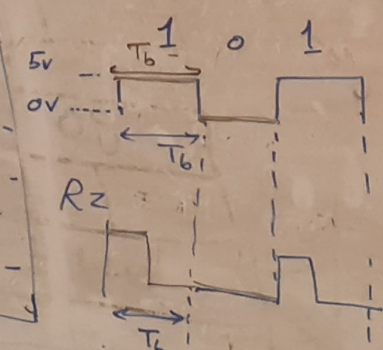
$q = \frac{7 - 6.7}{8}$

$q = 0.3$

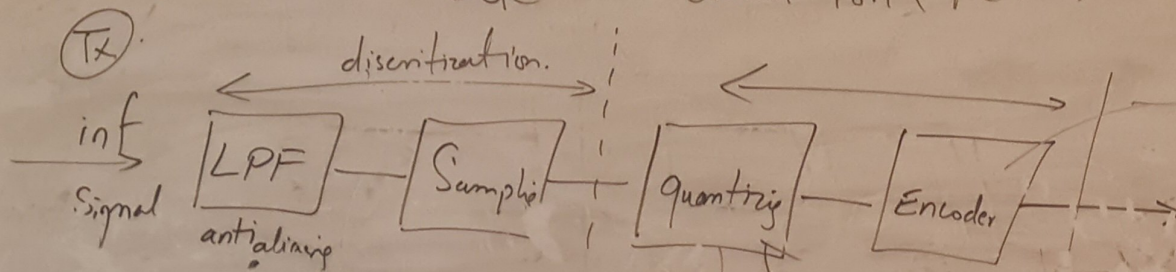


111 100 011

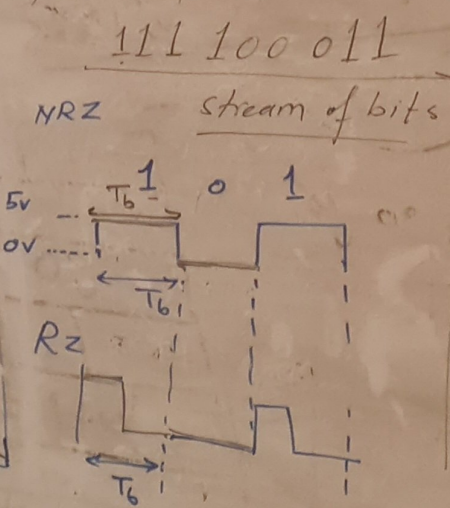
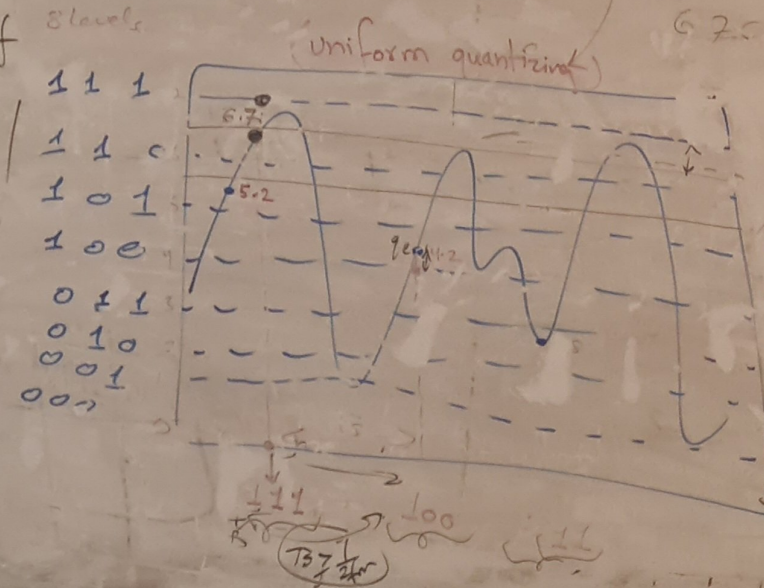
NRZ stream of bits



Pulse code Modulation (PCM)

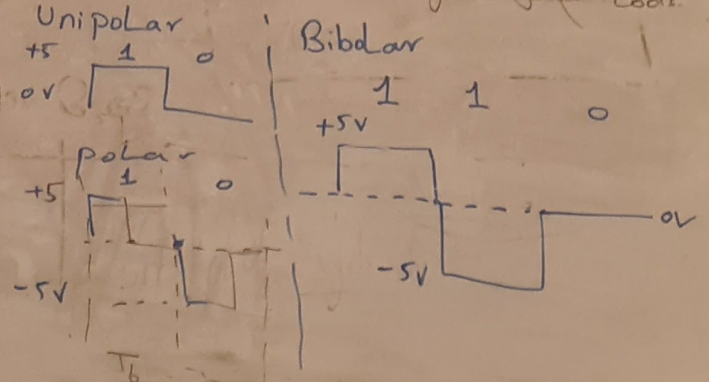


$q = \frac{7 - 6.7}{8} = 0.3$
 $q_c = 0.3$

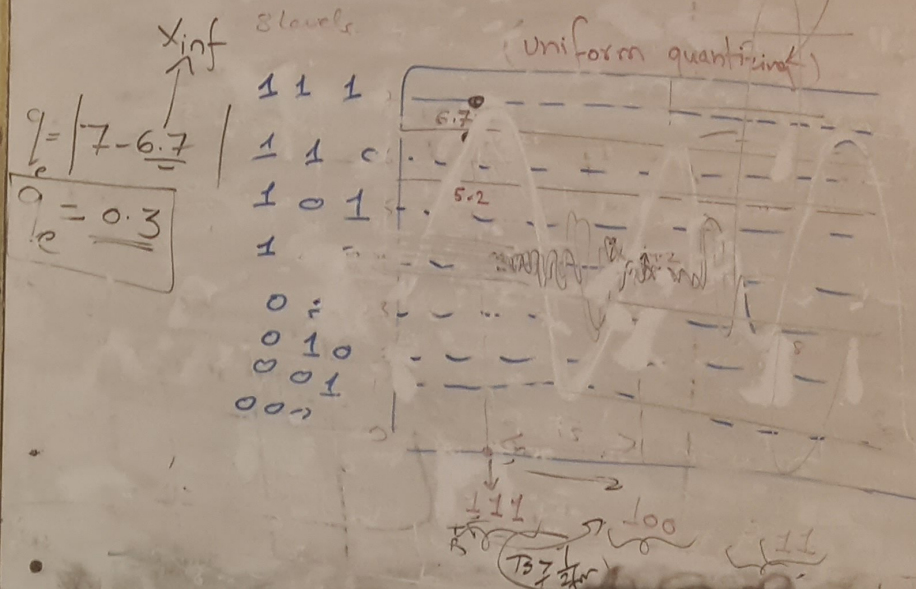
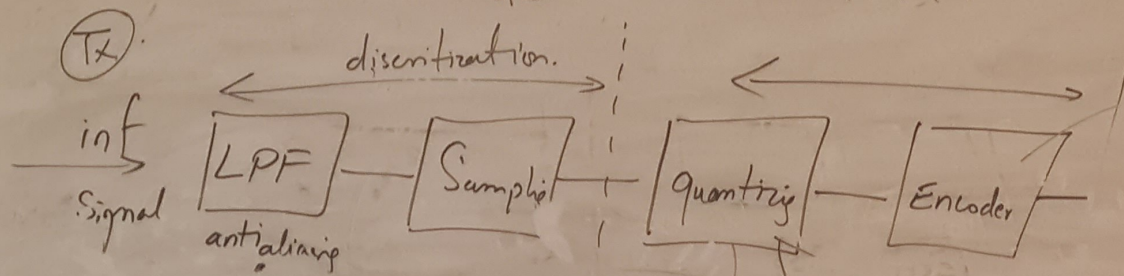


Line codes used for representing binary data stream

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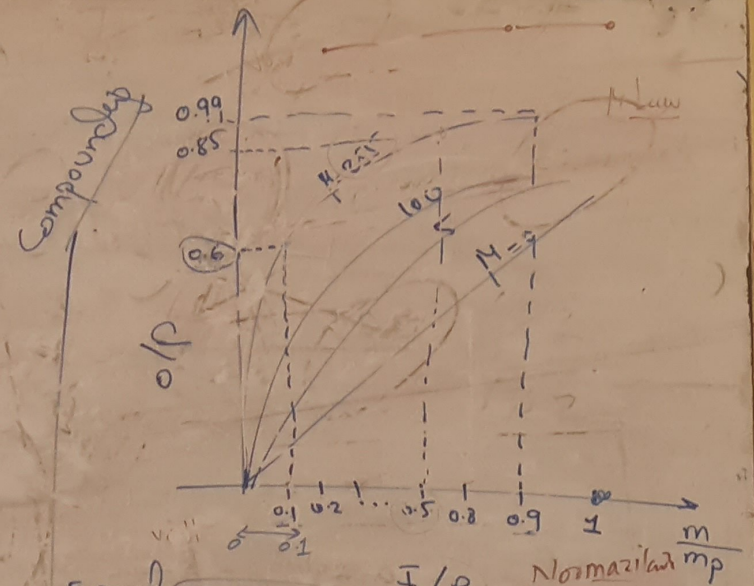
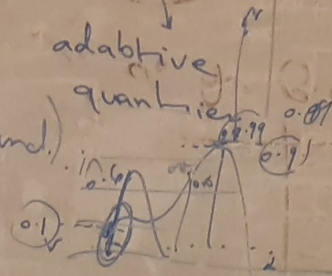
Pulse code Modulation (PCM)



Non-uniform quant

Companding
(compression + Expansion)

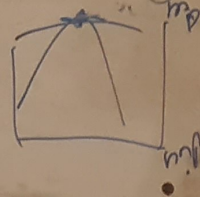
adaptive quantization



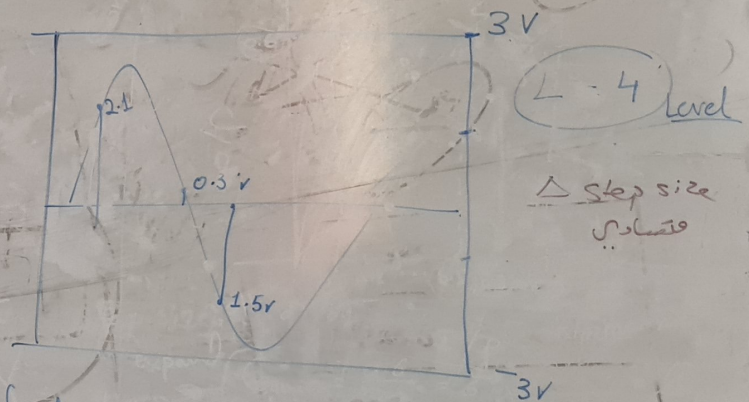
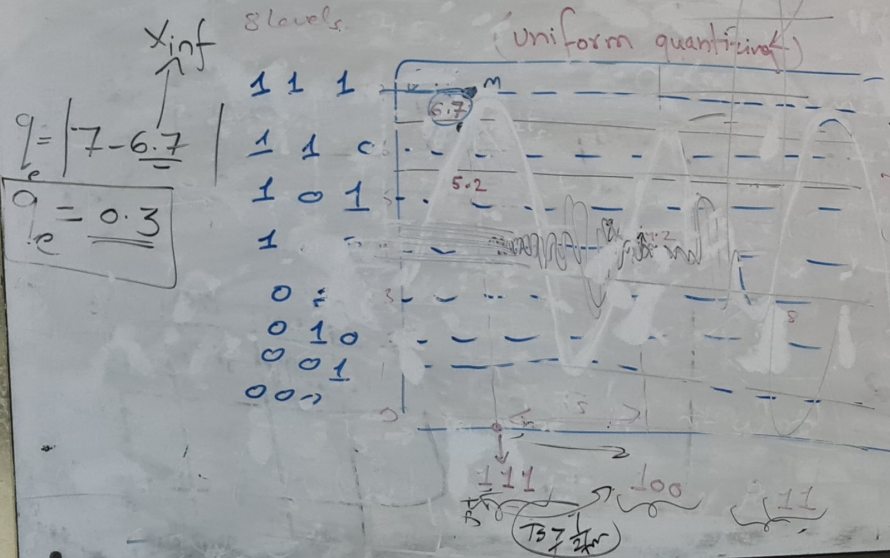
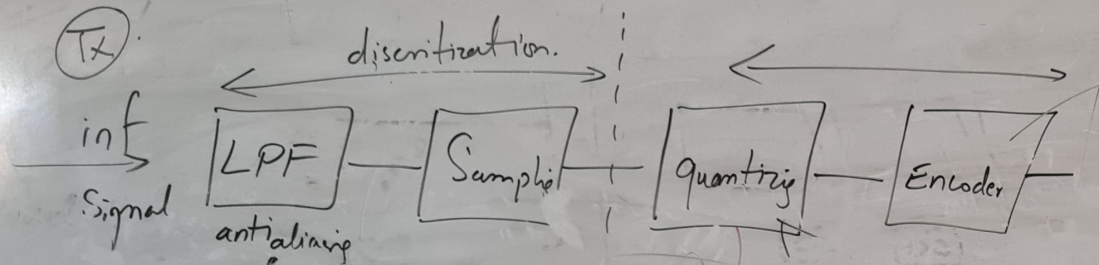
Expand $C_{gain} = \frac{0.6}{0.1} = 6$

$C_2 = \frac{85}{5}$, $C_1 = \frac{0.99}{0.9} = 1.1$

compression



Pulse code Modulation (PCM)



information sample	2.1V	0.3V	1.55V
quantized level	2.1V	0.3V	1.55V
quantization error			
line code			